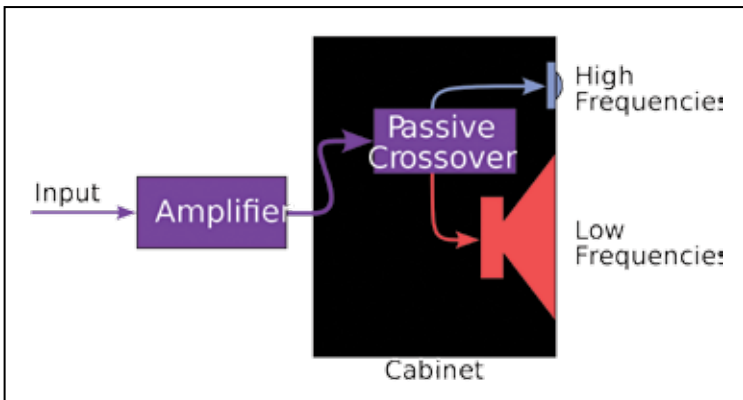


When a vendor specifies his product to be an active speaker, it's almost universally understood that they are referring to the audio crossover component in the system. If you remember the loudspeaker system consisting of multiple drivers within one loudspeaker system, you'll realise that the electrical signal needs to be routed to the appropriate driver, that is responsible for producing the corresponding frequency. Audio crossovers are the electronic filters split the audio signal into separate frequency bands that can be separately routed to the drivers optimized for those bands. In the present market, active crossovers are different from passive crossovers in that they divide the audio signal prior to amplification.

The main difference between passive and active speakers also lies in the design approach taken for the components used. Passive filters contain reactive components such as capacitors and inductors. Normally, cheaper passive speakers tend to contain individual components that don't function very well at high voltages. If you're looking to purchase very high performance passive crossover speakers, though, they are likely to be more expensive than active crossovers since individual components capable of good performance at the high currents and voltages at which speaker systems are driven are hard to make.

Another characteristic of some passive speakers is that they require an external amplifier to be connected externally. The signals received at the speaker are sufficiently powerful to render the sound without requiring any amplification. However, this means that the consumer requires a certain level of expertise to match the speaker with the right amplifier so that the



Passive speakers